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RECEIVED 18 November 2025

REVISED 29 December 2025

ACCEPTED 12 January 2026

PUBLISHED 30 January 2026

CITATION

Tsoku JT and Makatjane K (2026) Deep learning-based pairs trading: real-time forecasting of co-integrated cryptocurrency pairs. *Front. Appl. Math. Stat.* 12:1749337. doi: 10.3389/fams.2026.1749337

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Deep learning-based pairs trading: real-time forecasting of co-integrated cryptocurrency pairs

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Statistical arbitrage strategies, including pairs trading, rely on identifying co-movements and static long-term equilibrium relationships between assets, where conventional methods fail to capture non-stationary dynamics, hence reducing trading effectiveness. This study, therefore, addresses this challenge by employing a dynamic co-integration approach combined with deep learning techniques to select suitable cryptocurrency pairs and forecast spread dynamics. The study examines multiple cryptocurrencies, namely: BNB, Ethereum, Litecoin, Ripple, and USDT, using dynamic Johansen co-integration tests to identify pairs with time-varying equilibrium relationships, and model the spread through a Dynamic Weighted Ensemble of Deep Neural Network and Long Short-Term Memory. Forecasting accuracy, trading performance, and predictive uncertainty are evaluated using error metrics, trading outcomes, and 99% prediction intervals. The results indicate that only those cryptocurrencies with dynamically coherent relationships are suitable for mean-reversion strategies. Furthermore, the study found that the Dynamic Weighted Ensemble achieves the best predictive accuracy. At the same time, LSTM captures proportional temporal dynamics effectively, and the ensemble-driven trading signals generate timely buy and sell decisions with low-lag execution and robust management of market volatility. These findings, therefore, highlight the advantages of combining dynamic co-integration and adaptive deep learning for statistical arbitrage.

KEYWORDS

crypto prices, hybrid forecasting, neural networks, risk metrics, volatility

1 Introduction

Two major bear markets have affected global financial markets over the past two decades: the subprime mortgage Global Financial Crisis and the collapse of the high-technology bubble. These events exposed substantial limitations in conventional financial theories, as many investors experienced significant losses and were forced to delay retirement plans. In particular, heightened volatility and market unpredictability revealed the shortcomings of portfolio management techniques, such as Markowitz's mean-variance optimisation, in safeguarding investments against systemic risks. Consequently, market participants have increasingly questioned the reliability of these widely adopted methods and have shown growing interest in market-neutral long-short equity strategies that aim to minimize exposure to common risk factors while delivering relatively stable returns, even during periods of market stress [1].

Among such strategies, pairs trading has emerged as one of the most prominent and extensively studied market-neutral approaches. The strategy exploits temporary deviations in the price relationship between two assets by establishing offsetting long and short positions and profiting from subsequent convergence or divergence relative to an assumed equilibrium relationship [2]. Pairs trading gained early prominence in the 1980s through the work of Wall Street quantitative researcher Nunzio Tartaglia, who assembled a multidisciplinary team of mathematicians and computer scientists to develop computer-based trading systems that reduced human subjectivity in investment decisions [3, 4]. Although the strategy initially demonstrated strong performance, it subsequently experienced sustained periods of underperformance, highlighting its sensitivity to evolving market conditions [5]. The strategy achieved widespread academic recognition following the influential study of Gatev et al. [3], which formalized pairs trading through the introduction of the distance method (DM). Since then, pairs trading has become a foundational tool in quantitative finance and has been widely applied across different asset classes and market environments. Early empirical evidence suggests that the DM was historically profitable [6, 7]; however, subsequent studies document a deterioration in performance in more recent periods, particularly after accounting for transaction costs. Moreover, Grosu et al. [8] emphasized that profitability has been shown to vary significantly across market regimes, with greater resilience observed during periods of heightened volatility and financial stress. These mixed findings have motivated the exploration of alternative modeling approaches capable of capturing more complex dependence structures between asset prices.

In this context, copula and mixed-copula models have been proposed as flexible tools for modeling nonlinear and asymmetric dependence structures that correlation-based measures, such as Pearson correlation, fail to capture adequately [1]. While prior research has examined copula-based pairs trading strategies in specific markets and limited timeframes [9], comprehensive assessments remain scarce, particularly in large and rapidly evolving markets such as the Chinese equities market. To address this gap, Sun [1] conducted a systematic investigation using copula and mixed-copula methodologies on a diverse sample of Chinese stocks. Despite their flexibility, copula-based approaches have limitations. According to Flos et al. [10] their implementation requires specialized knowledge of copula theory and parameter estimation, making them more complex to specify, validate, and interpret than more conventional methods such as co-integration; accurate analysis of copula parameters is often data-intensive and typically relies on large volumes of high-frequency observations, which may be unavailable or costly to obtain [11]; the effectiveness of resulting trading strategies is highly sensitive to correct copula family selection and parameter calibration, with misspecification potentially leading to distorted tail-dependence characterization and unreliable trading signals; moreover, [12] emphasize that most standard copula models assume a static dependence structure, an assumption that is frequently violated in financial markets where correlations evolve, thereby motivating more complex and computationally demanding time-varying extensions; finally, copula models are largely statistical in nature and do not inherently provide an economic rationale for long-run co-movement between

assets, in contrast to co-integration frameworks that are grounded in equilibrium relationships. With respect to correlation analysis, it remains unable to identify long-term equilibrium relationships or adapt to changing market dynamics [13], while econometric techniques such as co-integration tests, error correction models, and Kalman filter approaches, although capable of modeling time-varying spreads, are constrained by assumptions of linearity and stationarity as indicated by Faizullin [14] and Eroğlu et al. [15]. In practice, estimation error, sensitivity to rolling-window selection, and parameter instability often undermine out-of-sample performance, and both correlation and copula-based methods face challenges when modeling high-dimensional interactions in rapidly evolving markets.

Recent studies published between 2023 and 2025 including studies by Rayaprolu [16], Rotondi and Russo [17], and Liou et al. [18] have sought to address these methodological limitations through enhanced statistical modeling, deep learning, and ensemble-based approaches. While these contributions have improved forecasting accuracy or volatility modeling, they typically focus on individual components of the pairs trading process, such as dependence modeling, spread construction, or prediction in isolation. In particular, recent ensemble learning applications, such as that of Ferrouhi and Bouabdallaoui [19], often rely on static or fixed-weight combinations that lack adaptability to regime changes, whereas econometric and copula-based methods remain limited in their ability to jointly capture nonlinear, high-dimensional, and time-varying dependencies. Among others, Keshavarz Haddad and Talebi [20] make a hypothetical portfolio composed of pairs of stocks by exploring a significant association between their prices in the Toronto Stock Exchange. The authors therefore compare the profitability of distance, co-integration, and copula functions as the pair's selection and trading strategy devices in TSX over January 2017 to June 2020. Building directly on these recent contributions, the present study advances the literature by integrating dynamic Johansen co-integration with deep neural networks (DNNs), long short-term memory (LSTM) networks, and a dynamic weighted ensemble (DWE) within a unified pairs trading setting. This integration enables the joint modeling of time-varying long-run equilibrium relationships, nonlinear and high-dimensional dependence structures, and evolving predictive relevance, capabilities that recent standalone statistical or machine learning approaches do not simultaneously provide.

A dynamic Johansen co-integration approach is employed in this study to identify cointegrated relationships among six major cryptocurrency pairs: Bitcoin–Ethereum (BTC–ETH), Bitcoin–Litecoin (BTC–LTC), Bitcoin–Ripple (BTC–XRP), Ethereum–Litecoin (ETH–LTC), Ethereum–Ripple (ETH–XRP), and Litecoin–Ripple (LTC–XRP). Unlike static co-integration approaches, which assume a fixed long-run equilibrium over the entire sample period, the dynamic formulation allows both the existence and strength of co-integrating relationships to evolve. This flexibility is particularly important in cryptocurrency markets, where rapid technological innovation, regulatory interventions, liquidity shifts, and speculative behavior frequently induce structural breaks and regime changes. By accommodating such temporal variation, the dynamic co-integration approach provides empirical support for the Adaptive Market Hypothesis (AMH)

of Lo [21] as cited by Kellner [22], which posits that market efficiency and asset interactions are inherently time-varying rather than static. Compared to static co-integration methods, dynamic co-integration reduces the risk of model mis-specification arising from unobserved structural shifts and avoids falsely rejecting long-run relationships that may hold intermittently rather than continuously. While correlation-based measures capture only short-term co-movements and are highly sensitive to volatility clustering, dynamic co-integration uncovers time-varying long-run equilibrium relationships even when individual price series are non-stationary [23, 24]. This enables the construction of adaptive mean-reverting spreads that remain economically meaningful across different market regimes, thereby providing a more robust foundation for market-neutral trading strategies and improved control of systemic risk [7].

These dynamically estimated spreads are subsequently used as inputs to Deep Neural Network (DNN) and Long Short-Term Memory (LSTM) architectures, allowing the models to learn from evolving equilibrium deviations rather than from static residual processes. Deep neural networks are well-suited to extracting complex nonlinear and high-dimensional patterns from these spreads, whereas LSTMs explicitly capture temporal dependencies and regime persistence, which are critical in highly volatile and non-stationary cryptocurrency markets [25, 26]. By integrating dynamic co-integration with deep learning models, the proposed approach aligns econometric theory with data-driven adaptability, thereby enhancing both predictive performance and economic interpretability.

Finally, this study introduces a dynamic weighted ensemble (DWE) that adaptively combines the complementary strengths of DNN and LSTM models. Unlike recent ensemble approaches that employ fixed see for instance [27, 28] or complexity-driven weighting schemes, the proposed DWE continuously reallocates model weights based on recent predictive performance, allowing the model combination to evolve as market conditions change. This adaptive mechanism enables the ensemble to respond effectively to regime transitions, structural breaks, and volatility spikes, thereby reducing out-of-sample forecasting errors during periods of rapid changes in model relevance.

1.1 Research highlights and key findings

This study assesses the predictive efficacy of deep learning models within a dynamic co-integration-based pairs trading framework, juxtaposing them with econometric models and introducing a dynamic weighted ensemble that adapts to variable market conditions. Consequently, Table 1 summarizes the principal findings of the research.

- Limitations of previous models: Prior studies on pairs trading predominantly employed Engle-Granger or Johansen co-integration methods in conjunction with linear mean-reversion signals [29–31], assuming stable statistical relationships that rarely persist in real markets marked by structural breaks and regime shifts.

TABLE 1 Research Key Findings.

Characteristic	Highlights	Key findings
Ensemble	Comparison of static ensembles (Bagging, Boosting, Stacking, Voting) with a Dynamic Weighted Ensemble.	DWE consistently outperforms static ensembles by adapting weights to real-time data conditions and regime shifts.
Dynamic weight adaptation	Incorporates time-varying performance metrics and market states into ensemble weighting.	Provides higher forecasting stability and accuracy, especially in volatile or structurally changing environments (e.g., cryptocurrency and financial markets).
Static ensemble limitations	Static ensembles apply fixed weights or model structures regardless of evolving data patterns.	Demonstrates reduced effectiveness during structural breaks, non-stationarity, or nonlinear dependencies—conditions where DWE excels.
Nonlinear learning capability	Uses deep learning models DNN and LSTM as base learners within the ensemble.	DWE enhances the non-linear modeling strengths of deep learning by prioritizing the best-performing model at each time step.
Regime sensitivity	Evaluates model behavior across different volatility regimes.	DWE adapts to regime shifts more effectively than Bagging, Boosting, or Stacking, leading to improved directional consistency and reduced forecasting error.
Error metric evaluation	Assessed using MSE, RMSE, MAE, MAPE, and MASE for robust comparison.	DWE achieves the lowest error values across metrics, confirming its superiority in high-frequency and non-stationary environments.
Computational efficiency	Efficient implementation using high-performance computing (HPC) and real-time systems.	Delivers faster convergence and reduced overfitting compared to Boosting and Stacking, where complexity grows rapidly with model depth.
Practical trading implications	Application to forecasting pair spreads for pairs trading strategies.	DWE provides more reliable trading signals, higher directional accuracy, and lower drawdowns than static ensemble approaches.

- Constraints of current machine learning ensembles: Recent applications of machine learning [32, 33] often employ fixed-weight or non-adaptive ensembles, which cannot adjust to evolving market dynamics, thus limiting their effectiveness in time-varying conditions.
- Dynamic co-integration method: This research utilizes a dynamic Johansen co-integration approach to ascertain time-varying equilibrium relationships among asset pairs, thus providing a more precise representation of market behavior.
- The dynamic weighted ensemble adjusts model contributions in real time based on current predictive performance, allowing for adaptation to changing market conditions and outperforming static ensemble methods.

- Complementary strengths of constituent models: The dynamic ensemble utilizes deep neural networks to manage nonlinearities, long short-term memory networks to capture temporal dependencies, and co-integration to maintain equilibrium-correcting behavior.
- Empirical advantage: The results demonstrate that the dynamic ensemble achieves higher predictive accuracy and greater stability in trading returns relative to static or fixed-weight ensembles.
- Alignment with the adaptive market hypothesis: This method aligns with Lo [21], emphasizing that model performance depends on variable market conditions and changing dependencies.
- Methodological and practical insights: Hybrid models utilizing adaptive weighting provide a robust framework for forecasting and exploiting co-integrated relationships within financial markets.

2 Methodology

This study establishes a pairs trading strategy specifically for cryptocurrencies, utilizing historical price data spanning from 02 January 2018 to 31 October 2025, and this gives a total of 2842 observations, which is obtained from <https://finance.yahoo.com/> using a yfinance package from Python accessed on 1st November 2025. Co-integrated pairs are identified using a dynamic Johansen co-integration method, which accounts for time-varying long-term equilibrium relationships, with six major cryptocurrency pairs considered: Bitcoin–Ethereum (BTC–ETH), Bitcoin–Litecoin (BTC–LTC), Bitcoin–Ripple (BTC–XRP), Ethereum–Litecoin (ETH–LTC), Ethereum–Ripple (ETH–XRP), and Litecoin–Ripple (LTC–XRP). The most suitable pairs for trading are selected based on their mean-reversion properties. The spread between each pair is modeled using DNN and LSTM networks to generate forecasts and predictive intervals, which support the formulation of dynamic trading signals. Model training and evaluation are conducted on high-performance computing (HPC) resources to handle the computational demands of deep learning, and the system is subsequently deployed for real-time prediction on cloud platforms AWS EC2 and Lambda, enabling automated, on-demand forecasting and trading signal generation.

2.1 Modeling assumptions and constraints

The proposed pairs trading in this study is developed under a set of explicit modeling assumptions designed to reflect the empirical properties of cryptocurrency markets while ensuring methodological tractability. First, individual cryptocurrency price series are assumed to be non-stationary, while linear combinations identified through dynamic Johansen co-integration are expected to exhibit mean-reverting behavior. This assumption underpins the construction of tradable spreads and is consistent with established statistical arbitrage literature.

Second, the long-run equilibrium relationships between asset pairs are assumed to be time-varying rather than static, motivating the use of a dynamic co-integration approach. This allows the

model to accommodate regime shifts, structural changes, and evolving dependence structures that are characteristic of high-volatility cryptocurrency markets. Third, it is assumed that the selected cryptocurrency pairs exhibit sufficient liquidity and market depth such that trading signals derived from the model can be executed without significant delays or excessive transaction cost distortions. While transaction costs are not explicitly modeled, the methodology proposed here is intended for highly liquid digital assets where such frictions are relatively limited. Fourth, deep learning architectures used in this study are assumed to be capable of capturing nonlinear, high-dimensional, and temporal dependencies in the co-integration-based spreads without imposing restrictive parametric distributional assumptions. The training procedure is designed to avoid look-ahead bias by strictly separating training, validation, and forecasting periods in a rolling and sequential manner. Finally, the dynamic weighted ensemble assumes that the relative predictive performance of individual models varies over time. Accordingly, model weights are updated adaptively based on recent forecasting accuracy, with the aim of improving robustness and reducing out-of-sample error during periods of rapid market change. These assumptions collectively define the scope and limitations of the proposed methodology and provide a transparent basis for interpreting the empirical results.

2.2 Deep neural network

A deep neural network is a type of artificial neural network with multiple hidden layers between the input and output layers. These hidden layers, according to Osgibemeh et al. [34], allow the network to learn complex patterns and representations from data, making them suitable for tasks such as image recognition, natural language processing, and time-series forecasting. Mathematically, the operation of a single neuron is expressed as follows

$$y_j = f\left(\sum_{i=1}^n w_{ji}x_i + b_j\right), \quad (1)$$

where x_i are the inputs ($i = 1, \dots, n$), w_{ji} are the weights connecting input i to neuron j , b_j is the bias for neuron j , $f(\cdot)$ is a nonlinear ReLU activation function, and y_j is the output of neuron j . While a single neuron captures only simple relationships, stacking multiple layers enables the network to model highly complex and nonlinear functions. In a DNN with L layers, the computations with respect to Awad and Khanna [35], proceed as follows

$$\mathbf{y}^l = f^l(\mathbf{W}^l \mathbf{y}^{l-1} + \mathbf{b}^l), \quad l = 1, \dots, L, \quad (2)$$

where $\mathbf{y}^0 = \mathbf{x}$ is the input vector. Here, $\mathbf{W}^l$ and $\mathbf{b}^l$ are the weights and biases for layer l , f^l is the ReLU activation function, and $\mathbf{y}^l$ is the output of the l -th layer.

Training a DNN involves adjusting the weights and biases across all layers to minimize a loss function, commonly using stochastic gradient descent (SGD), where the SGD performs a forward pass to make predictions, calculates the loss to assess error, and then executes a backward pass using backpropagation to compute gradients of the loss with respect to the model parameters.

These gradients are then used to update the weights, and the network's depth allows it to learn hierarchical representations from input data, with early layers capturing basic features and deeper layers learning more complex patterns [36]. In addition, Fan et al. [37] declared that DNNs are universal function approximators capable of automatically learning important features from raw input, making them widely used in areas such as image recognition, natural language processing, and time-series forecasting. Figure 1 illustrates the proposed DNN architecture used in this study.

2.3 Long short-term memory (LSTM) network

Long short-term memory networks are powerful for sequence modeling tasks such as time-series forecasting. The hidden layer's memory blocks constitute the LSTM's core, comprising cells and three gates: input, output, and forget gates. The forget gate f_t mitigates the limitations of standard recurrent neural networks by selectively forgetting irrelevant past information [38], while the input and output gates regulate information flow from the feature vector $\mathbf{x}_t$ to the hidden state $\mathbf{h}_t$. The computations within an LSTM cell are given by

$$i_t = \sigma(W_i \mathbf{x}_t + U_i \mathbf{h}_{t-1} + b_i), \quad (3)$$

$$\tilde{C}_t = \tanh(W_c \mathbf{x}_t + U_c \mathbf{h}_{t-1} + b_c), \quad (4)$$

$$f_t = \sigma(W_f \mathbf{x}_t + U_f \mathbf{h}_{t-1} + b_f), \quad (5)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (6)$$

$$o_t = \sigma(W_o \mathbf{x}_t + U_o \mathbf{h}_{t-1} + b_o), \quad (7)$$

$$\mathbf{h}_t = o_t \odot \tanh(C_t), \quad (8)$$

where i_t , f_t , and o_t are the input, forget, and output gate vectors; C_t is the cell state; $\tilde{C}_t$ is the candidate cell state; W_* and U_* are distinct weight matrices for each gate; b_* are bias vectors; σ and $\tanh$ are the sigmoid and hyperbolic tangent activation functions; and $\odot$ denotes the Hadamard (element-wise) product, defined as $(a \odot b)_i = a_i b_i$. The sigmoid function $\sigma(\cdot)$ maps inputs to $(0, 1)$, while $\tanh(\cdot)$ maps inputs to $(-1, 1)$, ensuring

stability in the cell state. The derivative of $\tanh(C_t)$ is bounded in $(0, 1)$, which in standard RNNs leads to vanishing gradients during backpropagation through time. The LSTM architecture mitigates this problem through the cell state C_t and gating mechanisms, which allow error signals to propagate over longer sequences without vanishing, preserving long-term dependencies. By explicitly using separate weight matrices for each gate and including gating mechanisms, the LSTM can effectively model complex temporal dependencies, even in volatile financial time series. Figure 2 illustrates the standard LSTM architecture and gating mechanisms.

2.4 Parameter estimation

Parameter estimation in this study is conducted in a structured and transparent manner to ensure statistical validity and reproducibility. Long-run equilibrium relationships among cryptocurrency prices are estimated using the Johansen co-integration procedure applied to log-transformed series. A parsimonious deterministic specification and fixed lag structure are adopted to balance model flexibility and stability, and the leading co-integrating vector is used to construct the equilibrium spread. This approach allows the estimation of co-integration parameters without imposing restrictive distributional assumptions while accommodating non-stationary price dynamics. For the deep learning models, parameters are estimated through supervised learning using rolling-window representations of the standardized co-integration spread. Both the deep neural network and long short-term memory models are trained via backpropagation using adaptive optimisation algorithms. Network weights and biases are iteratively updated by minimizing the mean squared error loss function, with convergence monitored through validation loss. Regularization mechanisms, including dropout, batch normalization, early stopping, and adaptive learning rate adjustment, are incorporated to stabilize estimation and mitigate overfitting. Temporal ordering is preserved throughout training to ensure that parameter estimates reflect genuine predictive structure rather than artifacts of look-ahead bias.

Input Layer

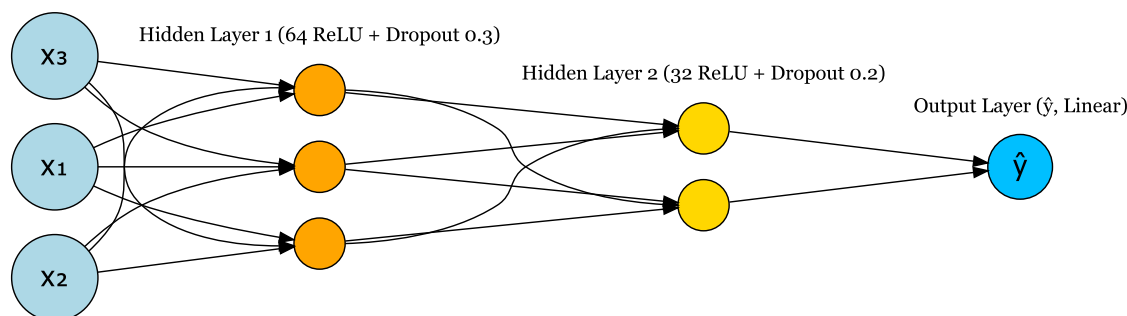
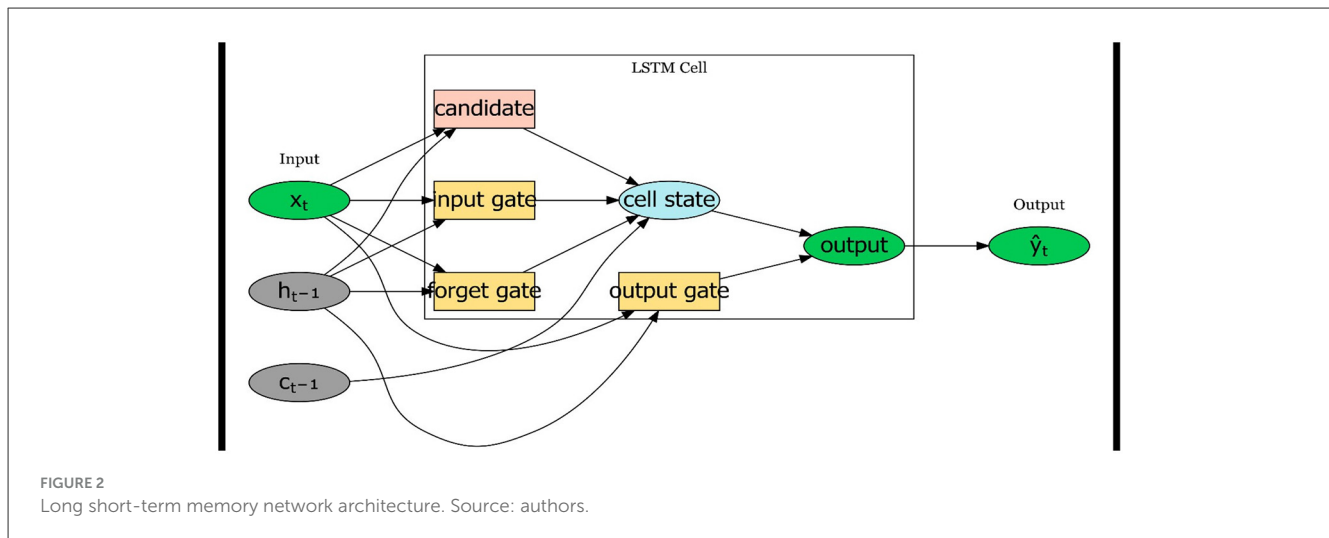


FIGURE 1
Deep neural network architecture. Source: authors.



The deep learning models, including the DNN and LSTM, are trained to minimize the mean squared error (MSE) between the predicted outputs and the observed values. Formally, the loss function is defined as follows

$$\mathcal{L} = \frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2, \quad (9)$$

where y_t is the true value at time t , $\hat{y}_t$ is the corresponding model prediction, and N is the number of observations in the training set. Training proceeds via backpropagation with adaptive optimisation algorithms, such as Adam, where weights and biases are iteratively updated to minimize $\mathcal{L}$. Regularization techniques, including dropout, batch normalization, early stopping, and learning rate scheduling, are incorporated to mitigate overfitting and stabilize convergence. For sequence models, temporal ordering is preserved to prevent look-ahead bias, ensuring that the loss minimization reflects genuine predictive performance.

2.5 Prediction interval widths and application to cryptocurrency spread forecasting

In predictive modeling for financial time series, it is crucial not only to produce accurate point forecasts but also to quantify the uncertainty associated with those forecasts. Prediction intervals (PIs) provide a probabilistic range around a point forecast, indicating where future observations are expected to fall at a specified confidence level [39]. The width of these intervals, known as the Prediction Interval Width (PIW), serves as a key measure of forecast uncertainty, with narrower intervals indicating greater confidence and wider intervals indicating higher uncertainty [40]. Parametric methods for constructing PIs often assume normality of forecast errors and static distributional properties; however, such assumptions are frequently violated in financial settings, where non-stationarity, heavy tails, and regime changes are common. To address this, this study employs

dynamic, distribution-free prediction intervals that adaptively adjust to changes in forecast uncertainty and do not rely on specific error distribution assumptions. These dynamic intervals are consistent with conformal prediction frameworks developed for time series, such as Temporal Conformal Prediction [41] and Bellman Conformal Inference [42], which construct well-calibrated intervals under distributional shifts and temporal dependence without assuming error normality. This contrasts with the Bayesian approach, which was used by Makatjane [43], which, while theoretically sound, is computationally intensive and sensitive to prior specification. For a forecast horizon $t+h$, let $\hat{y}_{t+h}$ denote the point forecast. The dynamic prediction interval is defined as

$$\text{PI}_{t+h} = \hat{y}_{t+h} \pm q_{\alpha/2}^{\text{dyn}} \cdot \hat{\sigma}_{t+h}, \quad (10)$$

where $\hat{\sigma}_{t+h}$ is the estimated forecast standard deviation, computed from the variance of an ensemble of model forecasts, and $q_{\alpha/2}^{\text{dyn}}$ is the empirical quantile of forecast errors corresponding to the desired confidence level. The Prediction Interval Width (PIW) is then computed by

$$\text{PIW}_{t+h} = 2 \cdot q_{\alpha/2}^{\text{dyn}} \cdot \hat{\sigma}_{t+h}, \quad (11)$$

where the factor 2 explicitly assumes asymmetric intervals around the point forecast. For the ensemble model, let $f_i(t+h)$ denote the forecast from the i -th base learner in an ensemble of M models. The ensemble mean forecast and variance are computed as follows

$$\hat{y}_{t+h}^{\text{ensemble}} = \sum_{i=1}^M w_i(t+h) f_i(t+h), \quad (12)$$

$$\hat{\sigma}_{t+h}^2 = \frac{1}{M} \sum_{i=1}^M (f_i(t+h) - \hat{y}_{t+h}^{\text{ensemble}})^2, \quad (13)$$

where $w_i(t+h)$ are normalized weights summing to 1; equal weights can be used as a simplification. The ensemble PIW is then given by

$$\text{PIW}_{t+h}^{\text{ensemble}} = 2 \cdot q_{\alpha/2}^{\text{dyn}} \cdot \hat{\sigma}_{t+h}. \quad (14)$$

The reliability and calibration of prediction intervals are evaluated using the following metrics:

Mean Prediction Interval Width (MPIW): the average width of prediction intervals over the evaluation period and it is computed as follows

$$\text{MPIW} = \frac{1}{N} \sum_{t=1}^N \text{PIW}_t, \quad (15)$$

Coverage Probability (CP): the proportion of actual observations lying within the prediction intervals. The indicator function is defined logically, without braces, as follows

$$\text{CP} = \frac{1}{N} \sum_{t=1}^N \mathbf{1} \left(y_t \in [\hat{y}_t \pm q_{\alpha/2}^{\text{dyn}} \hat{\sigma}_t] \right). \quad (16)$$

Minimum and Maximum PIW: reports the range of interval widths, highlighting periods of low or high predictive uncertainty, particularly during market stress. It is noted that CP calculations assume independence of forecast intervals over time. Dynamic, distribution-free methods, such as conformal prediction, explicitly account for temporal dependence and regime shifts, providing robust coverage even in non-stationary, volatile markets.

3 Results

Figure 3 illustrates the closing prices of five cryptocurrencies: Ethereum (black), BNB (blue), Litecoin (green), Ripple (red), and USDT (purple) from 02 January 2018 to 31 October 2025. Ethereum demonstrates significant valuation and volatility, with prominent price increases nearing 5,000 during the major bull cycles of 2021–2022 and 2024–2025. BNB has become the second-most valuable asset, demonstrating significant growth since 2021, with anticipated values exceeding 1,000 by 2024. In contrast, Litecoin, Ripple, and stablecoin USDT exhibit relatively stable price movements, with USDT largely adhering to its expected peg, in alignment with its function as a stablecoin. In addition to individual price dynamics, Figure 3 illustrates clear co-movement patterns among the assets.

Ethereum and BNB exhibit significant synchronous fluctuations during market-wide rallies and corrections, suggesting that these assets are responsive to broader cryptocurrency market cycles. Litecoin and Ripple show a weaker but still noticeable correlation with major assets, especially when the market is more volatile. These patterns demonstrate significant variability in price behavior and inter-asset relationships. Ethereum and BNB significantly influence market fluctuations, while Litecoin, Ripple, and USDT have a relatively minor impact. The observed dynamics warrant a focused analysis on Ethereum and BNB in the context of dynamic co-integration and pairs trading, as their significant price movements are expected to yield more statistically reliable signals for mean-reversion strategies.

3.1 Dynamic co-integration analysis for cryptocurrency pairs

Identifying cointegrated cryptocurrencies is essential for developing effective pairs trading strategies, as co-integration signifies a stable long-term equilibrium relationship conducive to mean-reversion trading. This study uses a dynamic co-integration methodology, as outlined by Franses [44], to address evolving market dynamics, differing from the static approach employed by Chen and Alexiou [45]. This approach facilitates the modeling of time-varying interactions among cryptocurrencies, thereby enabling more flexible and adaptive trading strategies in response to fluctuating market conditions. Panel (a) of Table 2 displays the initial dynamic cointegrating vector. USDT is identified as the dominant element of the non-stationary combination, exhibiting a coefficient of -270.904 , whereas Ethereum, BNB, Ripple, and Litecoin display relatively minor contributions. The cointegrating vector is used to create a dynamically non-stationary spread, which serves as the foundation for statistical arbitrage and mean-reversion trading. The eigenvalues of the co-integration matrix reinforce this conclusion: the first eigenvalue (0.0773) significantly exceeds the following values, indicating the existence of a predominant non-stationary combination. Panel (b) of Table 2 presents the time-varying lag Johansen trace statistics. The null hypothesis of no

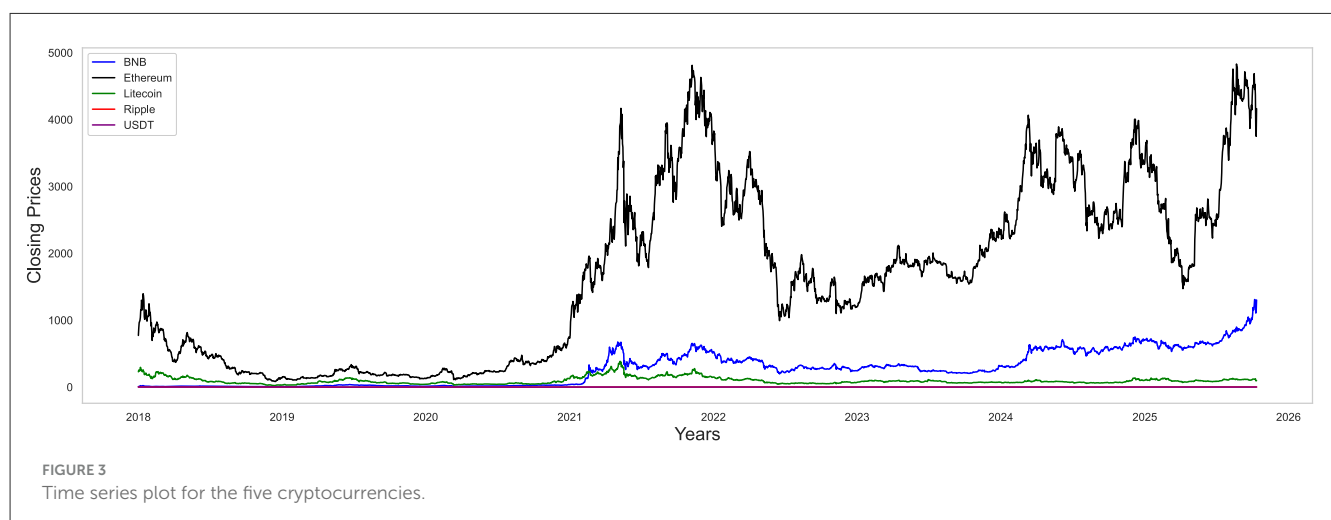


TABLE 2 Dynamic co-integration analysis.

(a) Dynamic co-integrating vector				
Crypto	Coefficient	Absolute coefficient		
USDT	−270.904	270.904		
Ethereum	−0.432	0.432		
BNB	0.124	0.124		
Ripple	0.067	0.067		
Litecoin	−0.030	0.030		
(b) Johansen trace test results (dynamic co-integration)				
Null Hypothesis (<i>r</i>)	Trace statistic	90% CV	95% CV	99% CV
<i>r</i> ≤ 0	276.319	65.820	69.819	77.820
<i>r</i> ≤ 1	47.799	44.493	47.855	54.681
<i>r</i> ≤ 2	19.299	27.067	29.796	35.463
<i>r</i> ≤ 3	4.832	13.429	15.494	19.935
<i>r</i> ≤ 4	0.537	2.705	3.841	6.635

co-integration ($r \leq 0$) is rejected at the 95% confidence level, as evidenced by a Trace statistic of 276.319, which surpasses the critical value of 69.819. The rejection of this hypothesis indicates the presence of a statistically significant co-integration relationship between USDT and the other four cryptocurrencies. The results indicate that the dynamically computed spread effectively captures a significant long-term equilibrium relationship, thereby validating its application in adaptive mean-reversion trading strategies. This approach explicitly models time-varying co-integration, ensuring that the trading strategy adapts to structural changes in cryptocurrency market behavior rather than depending on static measures.

3.2 Pairs trading strategy: a deep learning approach

The subsequent phase in executing the pairs trading strategy involves calculating the spread between the co-integrated cryptocurrencies. The hedge ratio is derived from the first co-integrating vector identified through the dynamic lagged Johansen test, reflecting the dynamic co-integration relationship among the selected assets, rather than through a static linear regression approach. The spread is defined as the difference in the logarithmic prices of the assets, adjusted by the co-integration coefficients and the dynamic score. This formulation identifies deviations from long-term equilibrium, serving as the basis for predictive modeling and the creation of trading signals. In contrast to Ko et al. [24], which evaluated six statistical methods for spread analysis, the current study utilizes DNN and LSTM models to analyse the computed spread. These models are well-suited for identifying potential mean-reversion opportunities, which are essential for developing effective trade signals. This modeling strategy is reinforced by Sun et al. [46], who conducts a systematic

survey of machine learning, deep learning, reinforcement learning, and deep reinforcement learning methods for pairs trading, thereby establishing a methodological foundation for replication and further advancement. Figure 4 presents the calculated spread from January 2018 to October 2025. The series initiates at a value close to zero and progressively decreases until the conclusion of 2021, indicating a substantial negative divergence from equilibrium. The spread demonstrates significant peaks during major market rallies, accompanied by abrupt declines and heightened volatility periods. From 2022 to late 2025, the spread exhibits fluctuations characterized by both upward and downward trends, along with periodic deviations exceeding the ± 1 standard deviation. The observed dynamics indicate periods when the market might be overvalued or undervalued relative to its long-term equilibrium, highlighting potential opportunities for mean-reversion. According to Monge et al. [47], these patterns can be used in a data-driven manner to produce actionable trading signals for cryptocurrency pairings.

The comparative evaluation of the DNN, LSTM, and proposed Dynamic Ensemble models across various forecasting error metrics is presented in Table 3. The Dynamic Ensemble demonstrates superior performance compared to individual models in magnitude-based forecasting, attaining the lowest MSE of 0.012124, RMSE of 0.110108, and MAE of 0.083607. The results demonstrate that the ensemble achieves the lowest absolute deviations between predicted and actual spread values, resulting in more precise point forecasts compared to the DNN or LSTM individually. The ensemble further shows the lowest mean forecast error (MFE) of -0.043546 , indicating minimal systematic bias and reinforcing its predictive reliability. The findings align with [48, 49], which indicate that ensemble methods outperform individual base learners. The LSTM demonstrates the lowest Mean Absolute Percentage Error (MAPE) of 1.490429%, indicating its proficiency in accurately capturing relative, percentage-based deviations and the temporal dependencies present in time series data. Although the magnitude-based errors of the LSTM exceed those of the DNN and ensemble methods, they demonstrate notable effectiveness in modeling proportional changes, which is crucial for dynamic trading strategies. The DNN exhibits strong performance in magnitude-based metrics, achieving the lowest Theil's U statistic (0.371179). Such behavior indicates improved efficiency compared to a naïve benchmark and demonstrates reliable scaled forecasting performance. The results highlight the complementary strengths of the three models: the DNN excels at benchmark-relative forecasting, the LSTM offers precise proportional predictions, and the dynamic ensemble combines these advantages to attain optimal accuracy and minimize bias.

Figure 5 illustrates the forecasting performance of the DNN, LSTM, and Dynamic Ensemble models during the test period. The ensemble model generates consistent predictions that align closely with the actual spread, while also preserving sensitivity to notable market fluctuations. The LSTM forecasts demonstrate increased variability, responding significantly to temporary market disruptions and swift structural alterations. The LSTM demonstrates a low MAPE; however, the heightened volatility may compromise the reliability of its predictions for trading decisions when used independently. The observed differences highlight



TABLE 3 Performance comparison of DNN, LSTM, and dynamic ensemble.

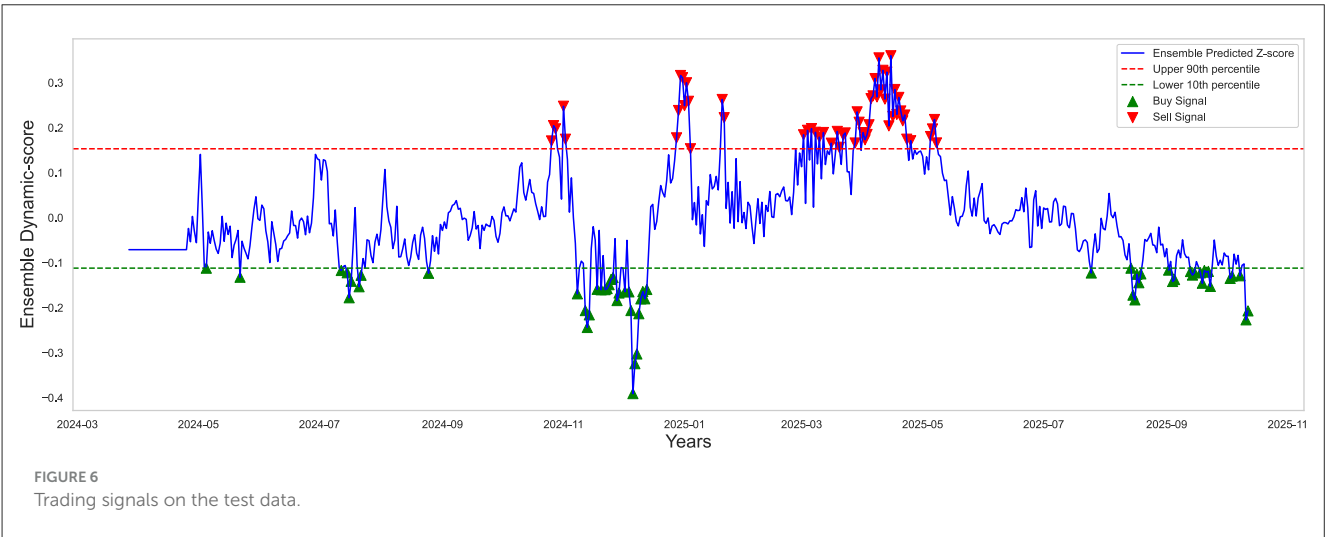
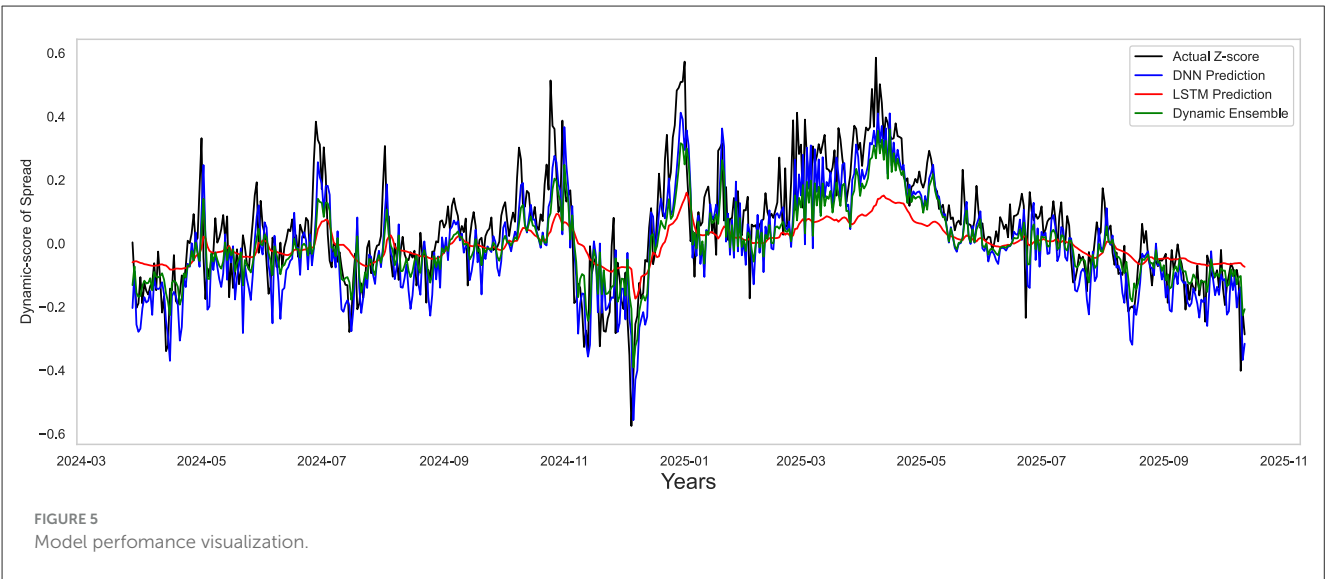
Metric	DNN	LSTM	Dynamic ensemble	Best model
MSE	0.017667	0.019226	0.012124	Dynamic ensemble
RMSE	0.132917	0.138658	0.110108	Dynamic ensemble
MAE	0.101259	0.104951	0.083607	Dynamic ensemble
MAPE (%)	4.098900	1.490429	2.033504	LSTM
MFE (Bias)	−0.052819	−0.044590	−0.043546	Dynamic ensemble
Theil's U	0.371179	0.794121	0.383306	DNN

the complementary characteristics of the models. The Dynamic Ensemble integrates the strengths of DNN and LSTM, resulting in enhanced accuracy and stability in predictions. This combination improves the reliability of trading signals obtained from the spread, thus facilitating more effective risk management and enhancing risk-adjusted performance in pairs trading strategies. The results underscore the significance of employing ensemble-based methods in cryptocurrency markets, characterized by nonlinear interactions and rapid temporal volatility, to ensure a balance between predictive precision and resilience to extreme fluctuations.

Following the demonstration that the Dynamic Ensemble surpasses the base learners (DNN and LSTM), the subsequent phase involves the creation of pairs trading signals. Figure 6 presents the dynamic ensemble forecast score alongside the associated trading signals. The signals originate from the predicted dynamic score, which quantifies deviations from the long-term equilibrium of co-integrated cryptocurrency pairs. The figure illustrates the upper and lower percentile thresholds, accompanied by a zero reference line denoting the mean. Trading decisions rely on scores in relation to specific thresholds: a Buy Signal is activated when the score drops below the lower threshold, suggesting a statistically undervalued state likely to increase, while a Sell Signal is generated when the score surpasses the

upper threshold, indicating overvaluation and a likely downward adjustment. In contrast to moving average-based trading methods [50], which presuppose relatively stable price behavior and struggle to account for time-varying dynamics, the proposed approach employs statistical deviations from the dynamically calculated spread. This facilitates adaptive trading that adjusts to fluctuating market conditions, especially in volatile cryptocurrency markets. Throughout the analyzed period, the algorithm produced 113 signals, comprising 81 winning trades (71.68%) and 32 losing trades (28.32%), thereby illustrating the strategy's efficacy in recognizing profitable opportunities. In comparison, Carta et al. [51] utilize explainable artificial intelligence (XAI) techniques, such as support vector regression, LightGBM, and random forest models, to demonstrate that trading strategies enhanced by feature selection yield predictive signals with increased informational content and diminished noise. The risk-return analysis indicates that XAI-powered strategies consistently surpass baseline techniques. The present ensemble-based method attains a comparable objective by combining various neural architectures to produce dependable, adaptive trading signals, while explicitly considering time-varying dynamics in the spread.

Evaluating the spread-ensemble trading mechanism, which produces buy and sell signals upon the expected spread crossing designated upper and lower thresholds, is crucial for determining the reliability and robustness of these trading outcomes. Table 4 presents essential performance metrics that assess the effectiveness, risk profile, and overall impact of the ensemble system on the pairs trading strategy. The hit rate of 0.5821 signifies that the ensemble accurately identifies profitable opportunities in approximately 58% of instances, demonstrating a relatively reliable predictive ability. The average profit per trade of 0.0111 indicates that transactions generally produce positive returns, aligning with the anticipated outcomes for high-frequency or mean-reversion strategies. The maximum drawdown (MDD) of -0.2875 indicates the most substantial peak-to-trough decline in cumulative returns. Although significant, this figure remains within acceptable thresholds for statistically driven strategies. This conclusion is corroborated by Choi [52], who assert that



drawdown-aware methodologies can decrease turnover compared to benchmark strategies. The risk-adjusted performance is robust, evidenced by a Sharpe Ratio of 1.3662 and a Sortino Ratio of 1.1411. The Sharpe Ratio demonstrates that returns significantly surpass the risk-free benchmark in relation to overall volatility, whereas the Sortino Ratio indicates effective management of downside risk. The average position size of 0.5113 indicates a moderate level of exposure, showcasing prudent risk management and the avoidance of excessive leverage. The estimated signal lag of 0.0000 days indicates the ensemble’s responsiveness to real-time market developments. The observed negative correlation with the market (-0.6517) indicates that the ensemble trading strategy may function as a diversifying asset, with the potential to outperform broader market trends during periods of underperformance, as noted in Sigauke et al. [27]. Table 4 demonstrates that the ensemble approach produces statistically significant trading signals while achieving a favorable balance among profitability, risk management, and market responsiveness, thereby confirming its practical applicability in volatile cryptocurrency markets.

TABLE 4 Risk performance for trading signals.

Metric	Value
Hit rate (win/loss probability)	0.5821
Average profit/loss per trade	0.0111
Maximum drawdown (MDD)	-0.2875
Sharpe ratio	1.3662
Sortino ratio	1.1411
Average position size	0.5113
Estimated signal lag (days)	0.0000
Correlation with market	-0.6517

3.3 Real-time signal generation for statistical arbitrage and stability test

The generation of trading signals for the statistical arbitrage strategy is executed in real-time through the deployment of the

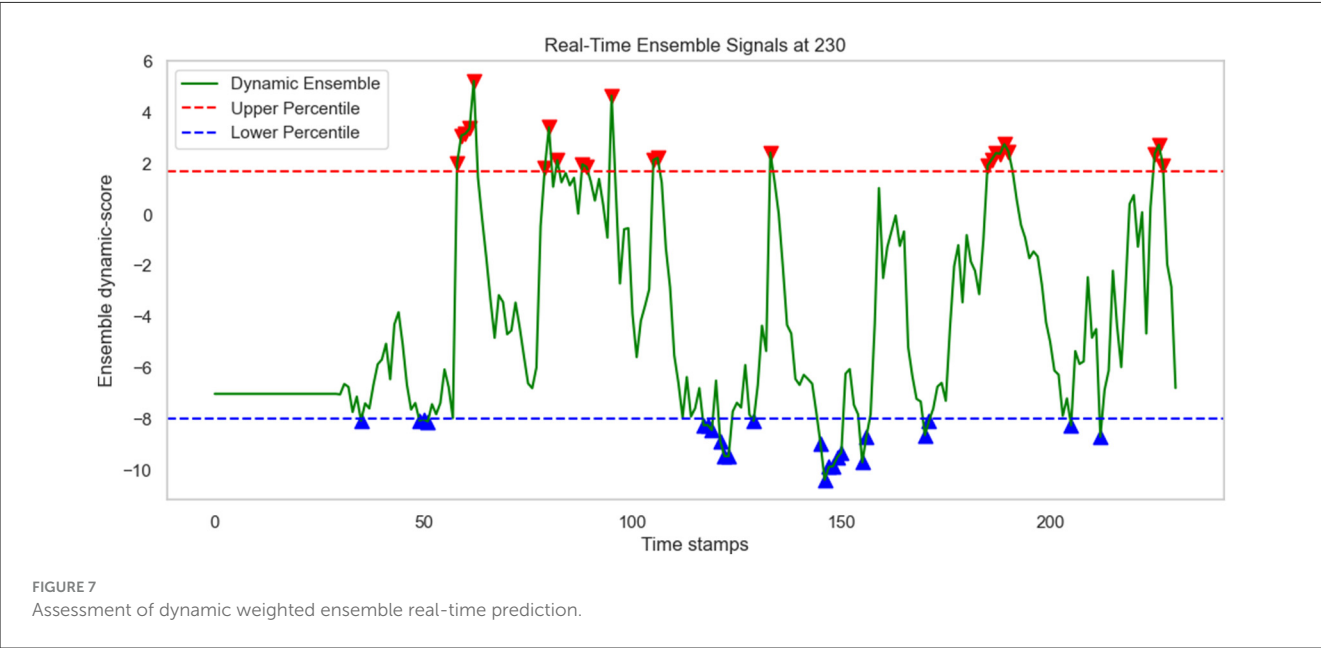
DNN-LSTM-ensemble on the AWS cloud platform, ensuring a scalable and operationally viable environment. The algorithm continuously acquires cryptocurrency price data from <https://www.binance.com/>, as illustrated in the Python script in Listing 1, and updates the dynamically calculated spread in real time. The dynamic score predicted by the ensemble is recalculated with each new observation, activating trade alerts upon threshold breaches. This configuration replicates a live market feed, facilitating the evaluation of the system’s responsiveness, accuracy, and robustness in streaming data scenarios. Table 5 and Figure 7 demonstrate the behavior of the ensemble in real time. Each entry documents the date index, the anticipated dynamic score, and the associated trading action—BUY, SELL, or HOLD. At Date 223, a notable negative deviation of -4.7369 is recorded; however, no trade is executed. This emphasizes the model’s dependence on empirically determined percentile thresholds rather than solely on absolute magnitude. The SELL signals are activated on Dates 225–227, with dynamic scores of 2.2797, 2.6762, and 1.8533, indicating statistically overbought conditions and a projected reversion to the

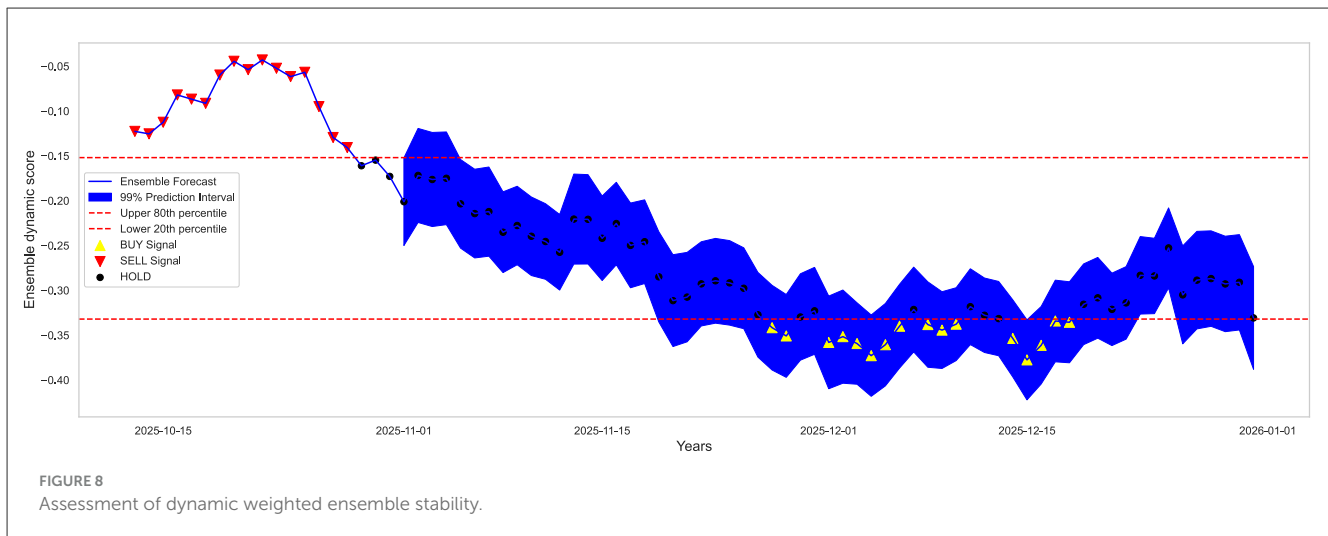
mean. In instances of intermediate or extreme negative deviations (e.g., Dates 223, 228–230), the model generates HOLD signals, reflecting a conservative strategy that reduces the likelihood of false positives. The results demonstrate that the ensemble system effectively differentiates actionable trading opportunities from transient volatility, facilitating precise and timely execution of statistical arbitrage strategies. The integration of real-time dynamic scoring with percentile-based thresholding improves the reliability of trading signals, thereby reinforcing the model’s practical applicability in active cryptocurrency markets.

Finally, the study assesses the predictive uncertainty of the DNN-LSTM-ensemble model, which is essential for evaluating its capacity to accurately forecast the spread between the chosen cointegrated cryptocurrency pairs and to ascertain its effectiveness in capturing potential deviations. The model’s 99% Prediction Interval Width (PIW) offers insights into the reliability and uncertainty of forecasts. During the test period, the average PIW is 0.0772, with a minimum value of 0.0232 and a maximum of 0.3094. This suggests that the model typically generates narrow intervals, demonstrating a high level of confidence in its predictions. The concluding PIW value of 0.0337 at the period’s end indicates that the ensemble sustains a relatively narrow prediction range, despite market fluctuations. The results indicate that the model effectively captures anticipated variations in the spread while adjusting to periods of heightened volatility. Figure 8 illustrates the predicted spread with the 99% confidence intervals represented in light blue. The observed spread values primarily align with these bands, validating the model’s capacity to represent both central tendencies and tail behavior. This alignment demonstrates the ensemble’s capacity to accurately characterize the distributional properties of the spread, including extreme deviations, which is crucial for risk-sensitive trading strategies. The stable and consistent out-of-sample forecasts generated by the DNN-LSTM-ensemble demonstrate its appropriateness for financial applications, where precise predictions of extreme movements and predictive reliability are essential. The findings align with Amnuaypongsa et al.

TABLE 5 Ensemble dynamic-scores and trading actions.

Date	Ensemble dynamic-score	Action
221	−1.3045	HOLD
222	0.0895	HOLD
223	−4.7369	HOLD
224	0.1206	HOLD
225	2.2797	SELL
226	2.6762	SELL
227	1.8533	SELL
228	−2.0775	HOLD
229	−2.8787	HOLD
230	−6.7983	HOLD





[53], which showed that penalizing larger prediction interval widths in multi-horizon forecasts enhances predictive confidence and decreases the probability of excessively broad intervals, despite their focus on renewable energy forecasting. The findings underscore the utility of the ensemble model in statistical arbitrage and dynamic pairs trading within volatile cryptocurrency markets.

4 Conclusion and recommendations

This study establishes a statistical arbitrage application driven by deep learning for dynamically co-integrated cryptocurrency pairs, demonstrating that predictive modeling significantly improves spread forecasting and signal reliability. The dynamic lagged Johansen procedure is employed to identify assets exhibiting non-stable long-run equilibrium behavior, as markets are not stable, especially the cryptocurrency, facilitating the construction of a spread that underpins forecasting and trade signal generation. The comparative analysis of the DNN, LSTM, and the proposed Dynamic Ensemble demonstrates distinct performance differences, where the Dynamic Ensemble demonstrates superior magnitude-based predictive accuracy, achieving the lowest MSE of 0.012124, RMSE, which is found to be 0.110108, and MAE of 0.083607, in addition to exhibiting the least forecast bias. The results support the benefits of combining complementary architectures for modeling nonlinear and temporally complex spread behavior. The LSTM, on the other hand, exhibits the lowest MAPE of 1.490429%, indicating its effectiveness in capturing proportional temporal dynamics, while the DNN shows benchmark-efficient forecasting with the lowest Theil's U value of 0.371179. These outcomes demonstrate that the models exhibit unique strengths: the LSTM is superior in relative percentage forecasting, the DNN is effective in benchmark-scaled scenarios, and the ensemble successfully integrates both attributes to achieve enhanced point-forecast accuracy and minimized bias; which [54] in their study also has found that combined forecasting using the stacking ensemble improves the accuracy of the base learners. The improved predictive capacity results in enhanced trading behavior, which is further evidenced by the ensemble's real-time signal accuracy, low-lag execution, and stable 99% prediction intervals, indicating that

most realized spread values are contained within the prediction interval bounds, reflecting effective uncertainty management, which is crucial in high-volatility cryptocurrency markets. These results are further reported by Makatjane and Shoko [55] in their study of explainable deep learning for financial risk. Though these authors examined the issue of underestimating tail risks in volatile cryptocurrency markets, particularly the Bitcoin/USD exchange rate, where models such as generalized autoregressive conditional heteroscedasticity (GARCH) frequently fail to predict extreme losses accurately, their exponentially smoothed recurrent neural network (ESRNN) at the 99% confidence level, effectively predicts the losses, indicating that the ESRNN yields dependable and precise Risk estimates in high-volatility markets, while with the current study, the findings indicate that deep learning, especially when integrated with dynamic weighting, significantly enhances the reliability and operational effectiveness of statistical arbitrage strategies.

Despite these impressive results, significant limitations remain. The empirical research looks at numerous cryptocurrency pairs, namely: BNB, Ethereum, Litecoin, Ripple, and USDT, but the conclusions are mostly limited to this group, which restricts generalisability until subsequently validated across more assets and marketplaces. Transaction costs, liquidity limits, and execution frictions are not explicitly represented; therefore, real profitability may vary from simulation results. The uncertainty analysis is limited to prediction interval widths and excludes higher-order tail-risk measures like Conditional Value-at-Risk (CVaR). Furthermore, the modeling structure does not include feature selection or explainable AI approaches, which could have increased interpretability and decreased structural noise. Therefore, future studies should incorporate a broader variety of cryptocurrencies and conventional asset classes to assess resilience across different market environments. Explainable AI techniques, as proposed by Carta et al. [51], will improve transparency and feature relevance when included, and this will enhance the model's capacity and prediction, and further indicate which features are more important in the prediction and their ability to influence the trading system, as markets are not stable in real time. Furthermore, including transaction-cost modeling, liquidity filters, and stress testing would improve the link

between theoretical performance and real execution. Investigating hybrid or transformer-based architectures, attention-augmented recurrent models, and reinforcement learning-based trading algorithms may also help to increase predicted accuracy and trading performance. Finally, including extreme-risk indicators in uncertainty quantification would allow for a more thorough evaluation of risk-return characteristics in statistical arbitrage applications, and all of these will be studied elsewhere.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

JT: Conceptualization, Validation, Writing – review & editing, Methodology, Project administration. KM: Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Conceptualization.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Acknowledgments

The authors are grateful to the numerous people who have provided helpful comments on this paper.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fams.2026.1749337/full#supplementary-material>

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Appendix

Listing 1 Python script for fetching real-time cryptocurrency prices from Binance API

```
1 from binance.client import Client
2
3 api_key = ""
4 api_secret = ""
5 client = Client(api_key, api_secret)
6
7 # Example: fetch current price
8 price = client.get_symbol_ticker(symbol=
9 "BTCUSDT")
10 print(price)
```